

## 2.12 Mass spectra and IR

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Students will be assessed on their ability to:

- a interpret fragment ion peaks in the mass spectra of simple organic compounds, eg the difference between propanal and propanone
- b use infrared spectra, or data from infrared spectra, to deduce functional groups present in organic compounds and predict infrared absorptions, given wavenumber data, due to familiar functional groups. This will be limited to:
  - i C-H stretching absorptions in alkanes, alkenes and aldehydes
  - ii O-H stretching absorption in alcohols and carboxylic acids
  - iii N-H stretching absorption in amines
  - iv C=O stretching absorption in aldehydes and ketones
  - v C-X stretching absorption in halogenoalkanes
  - vi as an analytical tool to show the change in functional groups during the oxidation of an alcohol to a carbonyl
- c demonstrate an understanding that only molecules which change their polarity as they vibrate can absorb infrared radiation
- d demonstrate an understanding that  $\text{H}_2\text{O}$ ,  $\text{CO}_2$ ,  $\text{CH}_4$  and  $\text{NO}$  molecules absorb IR radiation and are greenhouse gases, whilst  $\text{O}_2$  and  $\text{N}_2$  are not.